

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 7

INDIAN SPACE RESEARCH ORGANISATION - 5

INDIAN REGIONAL NAVIGATION SATELLITE SYSTEM (IRNSS)

Satellite Navigation:

- It is a system that uses satellites to provide position, navigation and tracking of objects fitted with a receiver.
- A satellite navigation system with a global coverage is termed as 'Global Satellite Navigation System'.
- Presently, the **GPS** of **USA**, **GLONASS** of **Russia** and **Galileo** of **European Union** are global systems.
- China is in the process of expanding its regional **BeiDou** navigation system into global by 2020.
- India, France and Japan have developed regional navigation systems.
- Global coverage can be achieved by a satellite constellation of 18-30 satellites in Medium Earth Orbit (MEO). Regional coverage can be achieved by 7-10 satellites.



INDIAN REGIONAL NAVIGATION SATELLITE SYSTEM (IRNSS)

- The Indian Regional Navigation Satellite System (IRNSS) is an independent satellite based regional navigation system developed indigenously by ISRO. It is also called 'NAVIC' (Navigation with Indian Constellation).
- The NAVIC system is constellation of eight satellites of which three are in geostationary and five are in geosynchronous orbit.
- The system covers entire India and a region extending 1,500 km area around the country's borders.



INDIAN REGIONAL NAVIGATION SATELLITE SYSTEM (IRNSS)

- This indigenous satellite-based navigation system is considered a deep strategic asset.
- The NAVIC system has both civilian and military uses.
- Moreover, it helps not just in land navigation but also in marine and aerial navigation.
- It offers wide services like
 - ✓ terrestrial and marine navigation
 - ✓ disaster management
 - ✓ vehicle tracking and fleet management,
 - ✓ visual and voice navigation for drivers.



MANGALYAAN: MARS ORBITER MISSION

- It is India's first interplanetary mission.
- It is an orbiter orbiting Mars since 24th September 2014.
- It was launched by ISRO on 5th November 2013.
- ISRO is the 4th space agency to reach Mars, after the Soviet Space Program, NASA and ESA (European Space Agency).
- MOM was launched using PSLV XL rocket.
- Through this mission India became the first Asian nation to reach Mars orbit and the first nation in the world to do it in the first attempt.



MANGALYAAN: MARS ORBITER MISSION

- The satellites development was completed in a record 15 months.
- The total cost of the mission was approximately 450 crores making it the least expensive Mars mission till date.
- The mission's primary objective is technology demonstration to develop technologies required for designing, planning, management and operations of an inter-planetary mission.
- The other objectives are to explore the Mars surface features, mineralogy and Martian atmosphere.



MANGALYAAN 2

- It is India's second inter-planetary mission planned for launch in the year 2024.
- It will consist of an orbiter and may consist of a lander and rover.
- The mission was announced in October 2014.
- In January 2016, India and France signed an agreement to jointly build Mangalyaan 2 by 2020.
- ISRO plans to use its heavy lifter GSLV Mk III for this mission.
- ISRO is still considering whether to have orbiter, lander and rover or only orbiter with better instruments.



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